

# Chern-Simons Gravity and Neutrino Self-Interactions

Cyril Creque-Sarbinowski  
Princeton/IAS Cosmo Lunch 2022



*arXiv: 2207.05094, Collaborator: Stephon Alexander*

# What is (dynamical) Chern-Simons Gravity [dCS]?

$$S = \int d^4x \sqrt{-g} \left[ \kappa R + \frac{a}{4f} {}^*R R - \frac{1}{2} (\nabla_\mu a) (\nabla^\mu a) \right]$$

↑

Spacetime  
Metric,  
Lorentzian  
Signature

↑

Einstein-  
Hilbert Term,  
 $\kappa = (16\pi G)^{-1}$

↑

dCS Decay  
Constant

↑

Pontryagin  
Density  
 ${}^*R R = {}^*R^\rho{}_\sigma{}^\mu{}_\nu R^\sigma{}_{\rho\mu\nu}$   
 ${}^*R^\rho{}_\sigma{}^\mu{}_\nu = \frac{1}{2} \epsilon^{\mu\nu\alpha\beta} R^\rho{}_{\sigma\alpha\beta}$

↑

dCS (pseudo)scalar

# What are Fermion Self-Interactions?

$$\mathcal{L}_\Psi = \bar{\Psi} (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi - \lambda \bar{\Psi} \Psi \bar{\Psi} \Psi$$

Diagram illustrating the components of the Fermion Lagrangian  $\mathcal{L}_\Psi$ :

- $i\gamma^\mu \partial_\mu$ : Dirac Gamma Matrices
- $\tilde{m}_\Psi$ : "bare" mass
- $\Psi$ : Dirac Fermion
- $\lambda$ : (Attractive) Fermion Self-Coupling Constant

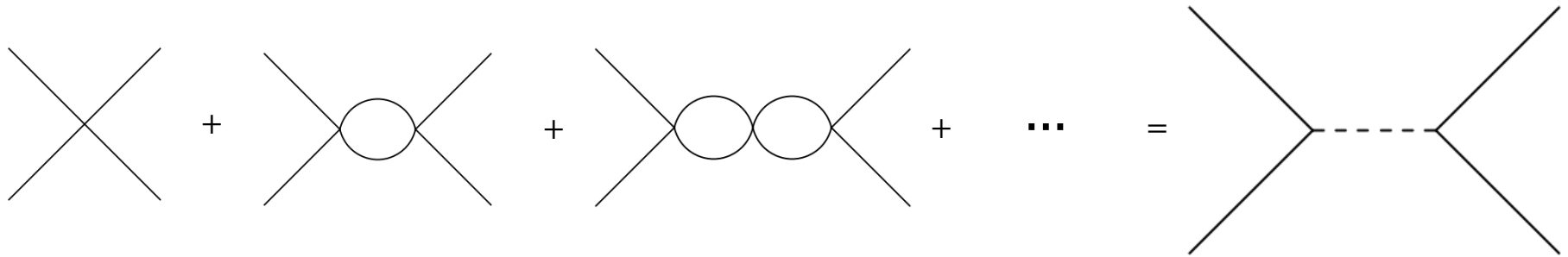
★ Cutoff  $\Lambda$

# How will we obtain dCS from Fermion-SI?

1. Identify and realize a scalar bound state in our theory of Fermion Self-Interactions (SI)
2. Demonstrate that the angular part of the (complex) scalar becomes the dCS scalar through fermion-loops with gravity after spontaneous symmetry breaking.

# How is there a bound state?

Geometric Summation to identify pole



# How to realize the bound state?

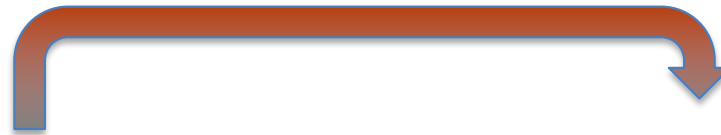
$$\begin{aligned}
 Z_\alpha &= \int \mathcal{D}\alpha \mathcal{D}\bar{\alpha} \left( - \int d^4x \tilde{m}_\Phi^2 \bar{\alpha} \alpha \right) \\
 &+ \\
 \Psi &= \Psi_\ell + \Psi_s \\
 \Phi &= \alpha - \tilde{m}_\Phi^{-2} \bar{\Psi}_s \Psi_s \\
 &+
 \end{aligned}
 \quad \Rightarrow \quad
 \begin{aligned}
 \tilde{\mathcal{L}}_\Psi &= \bar{\Psi} (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi - \lambda \bar{\Psi} \Psi \bar{\Psi} \Psi \\
 &+ (\partial_\mu \Phi^*) (\partial^\mu \Phi) - y_\Phi (\Phi \bar{\Psi} \Psi + \text{h.c.}) \\
 &+ m_\Phi^2 |\Phi|^2 - \frac{\lambda_\Phi}{4} |\Phi|^4,
 \end{aligned}$$

*Integrate Out Short – Scale Modes*

$$y_\Phi \sim \lambda \Lambda^2$$

# Spontaneous Symmetry Breaking

$$\Phi = (1/\sqrt{2})(F + \sigma) \exp[ia(x)/F]$$



$$\mathcal{L}_\Phi = \bar{\Psi}_L (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi_L + \bar{\Psi}_R (i\gamma^\mu \partial_\mu - \tilde{m}_\Psi) \Psi_R \\ + \partial_\mu \Phi \partial^\mu \Phi^* - V(|\Phi|^2) - (y_\Phi \Phi \bar{\Psi}_L \Psi_R + \text{h.c.}),$$

$$\mathcal{L}_\Phi = \bar{\Psi} \left[ i\gamma^\mu \partial_\mu - \left( \tilde{m}_\Psi + \frac{y_\Phi F}{\sqrt{2}} \right) \right] \Psi \\ + \frac{1}{2} \partial_\mu a \partial^\mu a - \frac{y_\Phi}{\sqrt{2}} a \bar{\Psi} \gamma^5 \Psi,$$



# The Gap Equation



$$\frac{\Delta m_\Psi}{m_\Psi} \left( \frac{2\pi^2}{\lambda\Lambda^2} \right) = 1 - \frac{m_\Psi^2}{\Lambda^2} \log \left( 1 + \frac{\Lambda^2}{m_\Psi^2} \right), \quad \Delta m_\Psi = y_\Phi F / \sqrt{2}$$

$\Lambda \gg m_\Psi$

$$F \approx 0.9m_\Psi$$

$\Lambda \ll m_\Psi$

$$F \approx 0.45\Lambda^2/m_\Psi$$

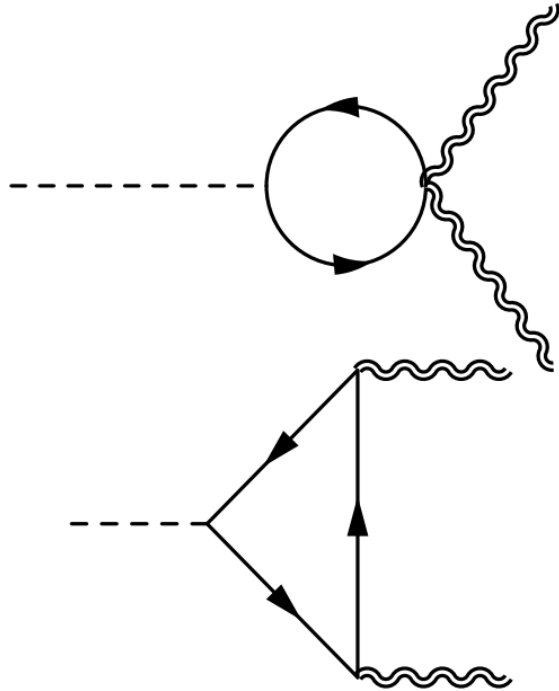
Remember!



$$y_\Phi \sim \lambda\Lambda^2$$



# Loop Generation of dCS

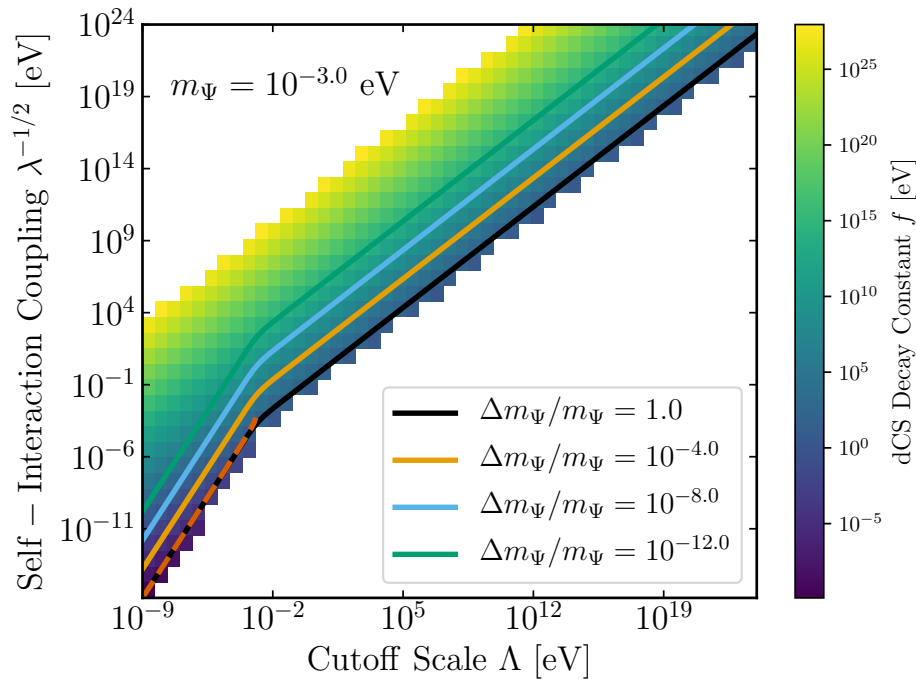


$$\mathcal{L}_g^{\text{eff}} = -\frac{g}{384\pi^2} \frac{a}{2m_\Psi} * RR$$

$$f = 192\pi^2 \frac{m_\Psi}{g}$$

$$g = y_\Phi/\sqrt{2}$$

# General Parameter Space



$$\Lambda \gg m_\Psi$$

$$f = 1.7 \text{ eV} \left( \frac{\Delta m_\Psi}{m_\Psi} \right)^{-1} \left( \frac{m_\Psi}{10^{-3} \text{ eV}} \right)$$

$$\Lambda \ll m_\Psi$$

$$f = 0.85 \text{ eV} \left( \frac{\Delta m_\Psi}{m_\Psi} \right)^{-1} \left( \frac{\Lambda}{m_\Psi} \right) \left( \frac{\Lambda}{10^{-3} \text{ eV}} \right)$$

# Example UV Completion for Fermion SI

(Scalar) Yukawa

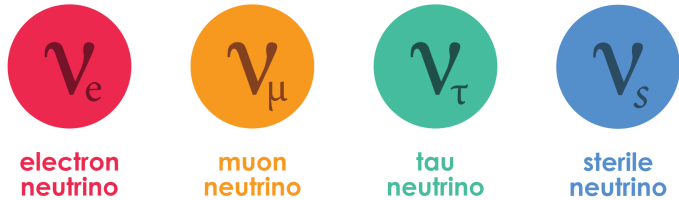
$$\mathcal{L} \supset g_\chi \chi \bar{\Psi} \Psi$$

$$\Lambda \sim m_\chi$$

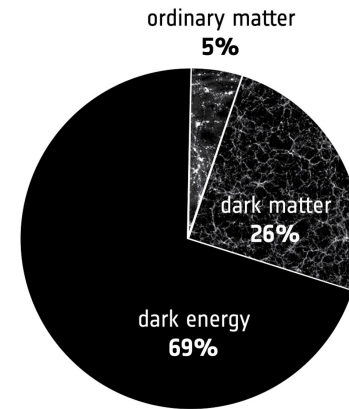
$$\lambda = (g_\chi / m_\chi)^2$$

$$f = 34 \text{ eV } g_\chi^{-2} \left( \frac{m_\Psi}{10^{-3} \text{ eV}} \right)$$

# Fermion Candidates



Source: <https://stringfixer.com/files/26342399.jpg>



Source:  
[https://cdn.sci.esa.int/documents/33220/35293/Cosmic\\_energy\\_energy\\_budget.jpg/840136ea-a65b-125d-e421-82d2ae889c06?version=1.0&t=1572356777315](https://cdn.sci.esa.int/documents/33220/35293/Cosmic_energy_energy_budget.jpg/840136ea-a65b-125d-e421-82d2ae889c06?version=1.0&t=1572356777315)

# Summary

1. If fermions self-interact, a complex scalar bound state forms
2. The bound state undergoes spontaneous symmetry breaking, giving a Yukawa term
3. The Yukawa term generates dCS through loops interactions with gravity
4. Light fermions, such as neutrinos (and possibly DM), are ideal candidates

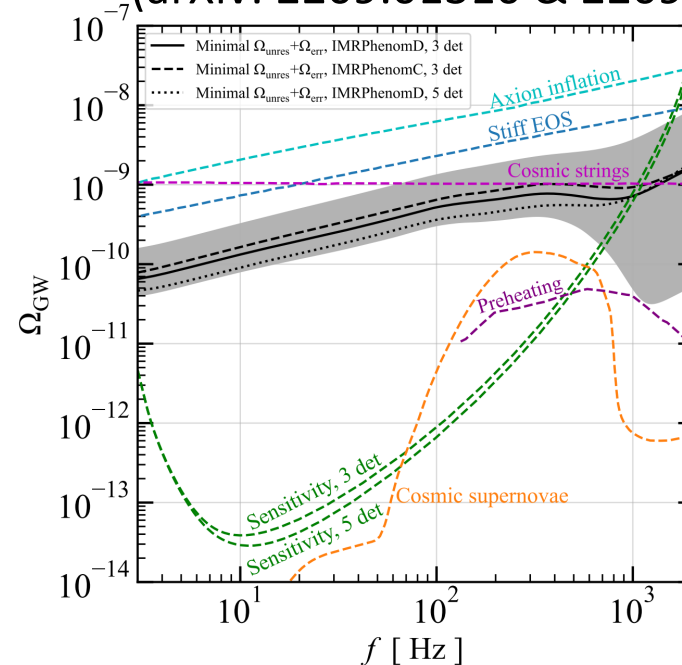
# Other works

Trispectrum from CSG

$$\left\langle \prod_{i=1}^4 \zeta(\mathbf{k}_i) \right\rangle$$

Collaborators: Stephon Alexander, Marc Kamionkowski

Subtracting Compact Binaries with XG Detectors  
(arXiv: 2209.01310 & 2209.01221)

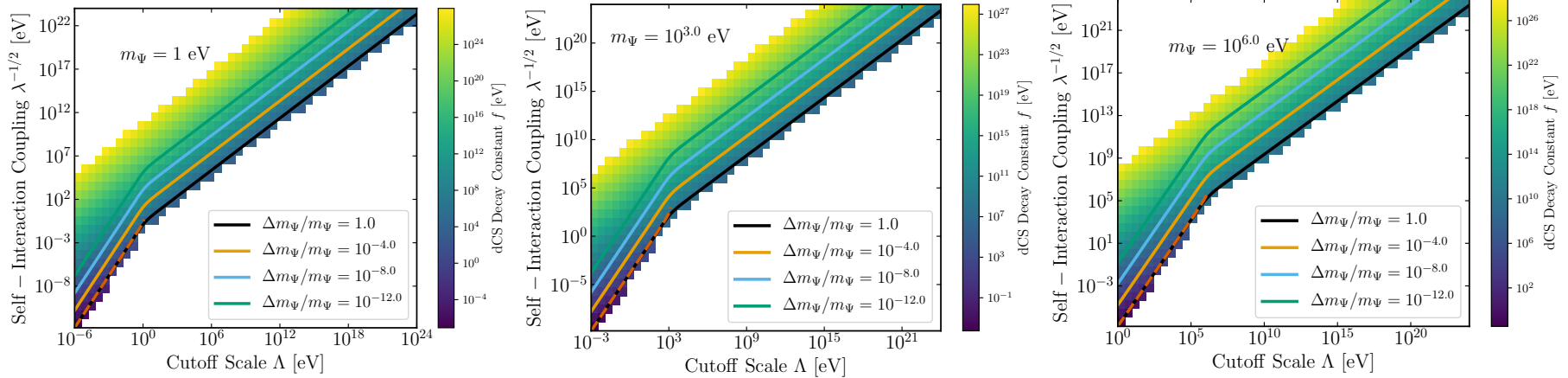


Authors: Bei Zhou, Luca Reali, Emanuele Berti, Mesut Çalışkan, Cyril Creque-Sarbinowski, Marc Kamionkowski, and B. S. Sathyaprakash



arXiv: 2207.05094, Collaborator: Stephon Alexander

# More Parameter Space Plots



# Exclusion Plots Neutrino Self-Interactions

